

Morning Paper Report - Friday, July 17, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Eccentricity Constrains Spatial Working Memory Fidelity: Evidence for the Cortical Maps Hypothesis

Siobhan M. McAteer, Anthony McGregor, Daniel T. Smith
Vision, 2026

Source: Crossref; matched terms: working memory representation, working memory, fidelity, precision, modelling, mnemonic precision

Link: <https://doi.org/10.3390/vision10030044>

Goal: Spatial working memory (SWM) has been characterised as a flexible resource that determines the precision with which memoranda are stored

Method: Mixture-modelling analyses indicated that the eccentricity effect was primarily driven by increases in imprecision at small set sizes (set sizes 1–5)

Results: Localisation error increased systematically with eccentricity, as predicted by the cortical maps hypothesis

Conclusion: The 'cortical map' proposal predicts that resource allocation, and therefore mnemonic precision, is limited by the availability of cortical space to represent memoranda

2. Behavioral and computational signatures of visual working memory deficits in adolescents with anxiety disorder

Tomoko Kishimoto, Ling Sun, Chongying Wang, Huayi Xu, Yudi Fu, Yu-Yan Gao, et al.
Current Psychology, 2025

Source: Semantic Scholar; matched terms: visual working memory, working memory, precision, computational model, computational models

Link: <https://www.semanticscholar.org/paper/33236866791f87212b14abeb36280c2363e1bfa4>

Goal: Pathological anxiety is one of the most common mental health problems in adolescents

Method: Using a computational psychiatry approach, we tested 14 computational models established in basic research of VWM

Results: Model comparison results identified the variable precision model as the best-fitting model for both groups, suggesting that the two groups share a qualitatively similar VWM process in completing the task

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3. A working memory model based on recurrent neural networks using reinforcement learning

Mengyuan Wang, Yihong Wang, Xuying Xu, Xiaochuan Pan
Cognitive Neurodynamics, 2024

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://www.semanticscholar.org/paper/950488c053df637c9092a6f953ba5eb422b87872>

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

4. Computational cognitive mechanisms of visual working memory in major depressive disorder and sex differences

Qingzu Kong, Qian Liu, Feng Gao, Xiang Wang, Zhiyan Wang, Chuman Xiao, et al.
BMC Psychology, 2025

Source: Semantic Scholar; matched terms: visual working memory, working memory, precision

Link: <https://www.semanticscholar.org/paper/abd2b56b3c9d83f2aadd144e291a53f4e32f9737>

Goal: This study examined the computational cognitive mechanisms of visual working memory (VWM) in MDD, focusing on memory precision while exploring potential sex differences

Method: The mainstream models of VWM were compared, and the variable-precision (VP) model was the best fit for our data

Results: Results revealed that MDD had worse memory precision than HC ($BF_{10} = 103.872$, decisive evidence for H1)

Conclusion: These findings highlight the critical role of diminished precision, specifically, larger resource allocation variability, in impaired VWM in MDD. Meanwhile, these findings highlight sex differences in memory precision and initial resources of VWM.

5. A Computational Cognitive Model of Human Memory Based on Invertible Neural Networks

Zheyuan Zhang
2022

Source: Crossref; matched terms: computational model, neural network, neural networks, human memory

Link: <https://doi.org/10.31219/osf.io/58adh>

Goal: We present a computational model of memory to simulate how humans remember and forget information based on the multi-store model of memory, which divides the memory into a sensory register, a short-term store, and a long-term store

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Results: The model recovers memory traces using invertible neural networks (INNs). We also use this model to explain some cognitive phenomena.

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